

Roll No. :

END SEMESTER EXAMINATION December 2024

Name of the Program: **B.Tech**

Semester: VII

Name of the Course: Artificial Intelligence

Course Code: TOE 714

Time : 3 Hours

Maximum Marks : 100

Note:

- i. **All questions are compulsory.**
- ii. Answer any two sub questions among a, b & c in each main questions
- iii. Total marks in each main question are twenty.
- iv. Each questions carries 20 marks

Q1	(20Marks)	
(a)	What are intelligent Agents? Discuss the properties of AI agents. How does an intelligent agent approach problem-solving?	CO1
(b)	Explain context-free grammar and its role in natural language parsing. Parse the sentence: "The cat sat on the mat" using a context-free grammar.	CO3
(c)	Derive a parse tree for the sentence "Bose loves the fish", where the following rules are used: $S \rightarrow NP VP$ $NP \rightarrow N$ $NP \rightarrow DET N$ $VP \rightarrow V NP$ $DET \rightarrow the$ $V \rightarrow loves$ $N \rightarrow Bose/fish$	CO5
Q2	(20Marks)	
(a)	Describe the concept of a semantic net in knowledge representation. Explain its use in NLP, and expert systems.	CO1
(b)	Check the validity. $P \rightarrow (Q \rightarrow R)$ equivalent to $(P \rightarrow Q) \rightarrow (P \rightarrow R)$	CO3
(c)	Design a simple semantic net to represent the relationship between the following entities: "John owns a dog, and the dog's name is Max."	CO4
Q3	(20Marks)	
(a)	What is the Turing Test in artificial Intelligence? Explain.	CO1
(b)	Define the problem as State Space Problem State and Prove Water Jug Problem (you have two jugs, a 5-liter and other 3 liter with no measuring marker on them. There is endless supply of water through tap. Our task is to get 4 liter of water in the 5 liter jug).	CO5

(c)	How are search strategies used for problem solving in AI? What is the idea of heuristics search in AI?	CC
Q4	(20Marks)	
(a)	Explain the significance of Artificial Intelligence in simulating intelligent behavior in different areas.	CO2
(b)	Define the following- i. Horn clause, ii. Goal clause, iii. Dual Horn clause, iv. Horn formula, v. Definite Clause.	CO1
(c)	Convert the following in First order predicate logic. 1. All birds fly. 2. Every man respects his parents.	CO4
Q5	(20 Marks)	
(a)	Define expert systems and give examples of existing systems like DENDRAL and MYCIN.	CO1
(b)	Discuss the depts.-first and breadth-first searching methods. Suggest some application to which each is best suited?	CO2
(c)	Write a PROLOG program to determine if a number is even or odd.	CO5